

Databases – Exercises Database Normalisation

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1. We consider a relation $R(A, B, C, D, E, F, G)$ with functional dependencies:

$$\begin{array}{llll} A \rightarrow E, F & A, B, C \rightarrow D, G & E \rightarrow C, G & C \rightarrow B, G \\ B, C \rightarrow E & A, B, C \rightarrow D & F \rightarrow D, E & E, G \rightarrow C, D \end{array}$$

- (a) Compute the cover of $\{C\}$.

Solution:

$$\{C\}^+ = \{B, C, D, E, G\}$$

- (b) Determine which of the functional dependencies

$$C \rightarrow A, F \quad C \rightarrow D \quad C \rightarrow B, D \quad C \rightarrow F \quad C \rightarrow A, C$$

is implied by the functional dependencies above.

Solution: The following two functional dependencies are implied:

$$C \rightarrow D \quad C \rightarrow B, D$$

2. We consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$A \rightarrow B \quad A, D, E \rightarrow B, C \quad B, E \rightarrow A \quad C \rightarrow B, E$$

Compute all minimal keys of the relation R .

Solution:

- The set of attributes that does not occur in any right-hand side is $\{D\}$.
- We start with candidates $\{\{D\}\}$.
- We consider a smallest candidate $\{D\}$: $\{D\}^+ = \{D\}$.
We extend the candidate with all attributes that are not in the cover:

$$\{\{A, D\}, \{B, D\}, \{C, D\}, \{D, E\}\}$$

- We consider a smallest candidate $\{C, D\}$: $\{C, D\}^+ = \{A, B, C, D, E\}$.
So, $\{C, D\}$ is a **minimal key**. The remaining candidates are

$$\{\{A, D\}, \{B, D\}, \{D, E\}\}$$

- We consider a smallest candidate $\{A, D\}$: $\{A, D\}^+ = \{A, B, D\}$.
We extend the candidate with all attributes that are not in the cover:

$$\{\{B, D\}, \{D, E\}, \{A, D, E\}\}$$

We do not need to consider $\{A, C, D\}$ since it contains a key we already found.

- We consider a smallest candidate $\{B, D\}$: $\{B, D\}^+ = \{B, D\}$.
We extend the candidate with all attributes that are not in the cover:

$$\{\{D, E\}, \{A, D, E\}, \{A, B, D\}, \{B, D, E\}\}$$

We do not need to consider $\{B, C, D\}$ since it contains a key we already found.

- We consider a smallest candidate $\{D, E\}$: $\{D, E\}^+ = \{D, E\}$.
We extend the candidate with all attributes that are not in the cover:

$$\{\{A, D, E\}, \{A, B, D\}, \{B, D, E\}\}$$

We do not need to consider $\{C, D, E\}$ since it contains a key we already found.

- We consider a smallest candidate $\{A, D, E\}$: $\{A, D, E\}^+ = \{A, B, C, D, E\}$.
So, $\{A, D, E\}$ is a **minimal key**. The remaining candidates are

$$\{\{A, B, D\}, \{B, D, E\}\}$$

- We consider a smallest candidate $\{B, D, E\}$: $\{B, D, E\}^+ = \{A, B, C, D, E\}$.
So, $\{B, D, E\}$ is a **minimal key**. The remaining candidates are

$$\{\{A, B, D\}\}$$

- We consider a smallest candidate $\{A, B, D\}$: $\{A, B, D\}^+ = \{A, B, D\}$.
We extend the candidate with all attributes that are not in the cover. We do not need to consider $\{A, B, C, D\}$ and $\{A, B, D, E\}$ since they contain a key we already found. So there are no more candidates to consider.

Thus the relation has 3 minimal keys:

$$\{C, D\}$$

$$\{A, D, E\}$$

$$\{B, D, E\}$$

3. We consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$D \rightarrow A \quad B, C \rightarrow D \quad A, B, C \rightarrow D, E \quad C \rightarrow D \quad B, E \rightarrow A, C$$

Compute a canonical set of functional dependencies.

Solution: First we make the right-hand sides singular:

$$\begin{array}{lllll} D \rightarrow A & B, C \rightarrow D & A, B, C \rightarrow D & C \rightarrow D & B, E \rightarrow A \\ & & A, B, C \rightarrow E & & B, E \rightarrow C \end{array}$$

Next, we remove implied functional dependencies:

- $B, C \rightarrow D$ is implied by $C \rightarrow D$,
- $A, B, C \rightarrow D$ is implied by $C \rightarrow D$, and
- $B, E \rightarrow A$ is implied by $B, E \rightarrow C$, $C \rightarrow D$ and $D \rightarrow A$.

This yields

$$A, B, C \rightarrow E \quad B, E \rightarrow C \quad C \rightarrow D \quad D \rightarrow A$$

Next we minimise left-hand sides:

- $A \in \{B, C\}^+$, so we can replace $A, B, C \rightarrow E$ by $B, C \rightarrow E$

Now we have

$$B, C \rightarrow E \quad B, E \rightarrow C \quad C \rightarrow D \quad D \rightarrow A$$

No more attributes can be dropped from any left-hand side.

Lets check whether we have more implied functional dependencies:

- the cover of $\{B, C\}$ without using $B, C \rightarrow E$ is $\{A, B, C, D\}$ (does not contain E)
- the cover of $\{B, E\}$ without using $B, E \rightarrow C$ is $\{B, E\}$ (does not contain C)
- the cover of $\{C\}$ without using $C \rightarrow D$ is $\{C\}$ (does not contain D)
- the cover of $\{D\}$ without using $D \rightarrow A$ is $\{D\}$ (does not contain A)

So there are no more implied functional dependencies.

Hence the canonical set of functional dependencies is:

$$B, C \rightarrow E \quad B, E \rightarrow C \quad C \rightarrow D \quad D \rightarrow A$$

4. We consider a relation $R(A, B, C, D, E)$ with **canonical** set of functional dependencies:

$$B, C, D \rightarrow E \quad A \rightarrow B \quad E \rightarrow B \quad E \rightarrow C \quad A, E \rightarrow D \quad A \rightarrow C$$

Use the synthesis algorithm to compute the Boyce-Codd normal form of R.

Solution: The first step of the synthesis algorithm is to compute a canonical set of functional dependencies. Luckily the given set is already canonical.

Next, we maximise the right-hand sides of the functional dependencies:

$$A \rightarrow B, C \quad A, E \rightarrow B, C, D \quad B, C, D \rightarrow E \quad E \rightarrow B, C$$

We now split R along violating functional dependencies:

- We start with $R(A, B, C, D, E)$.
- The functional dependency $B, C, D \rightarrow E$ violates the BCNF of R, so we split:

$$R(A, B, C, D) \quad R_2(B, C, D, E)$$

- The functional dependency $E \rightarrow B, C$ violates the BCNF of R_2 , so we split:

$$R(A, B, C, D) \quad R_2(D, E) \quad R_3(B, C, E)$$

- The functional dependency $A \rightarrow B, C$ violates the BCNF of R, so we split:

$$R(A, D) \quad R_2(D, E) \quad R_3(B, C, E) \quad R_4(A, B, C)$$

There is nothing more to split, so the BCNF is:

$$R(A, D) \quad R_2(D, E) \quad R_3(B, C, E) \quad R_4(A, B, C)$$

5. We consider a relation $R(A, B, C, D, E)$ with **canonical** set of functional dependencies:

$$B, C, D \rightarrow E \quad A \rightarrow B \quad E \rightarrow B \quad E \rightarrow C \quad A, E \rightarrow D \quad A \rightarrow C$$

Use the synthesis algorithm to compute the third normal form of R.

Solution: The first step of the synthesis algorithm is to compute a canonical set of functional dependencies. Luckily the given set is already canonical.

Next, we merge functional dependencies with the same left-hand side:

$$A \rightarrow B, C \quad A, E \rightarrow D \quad B, C, D \rightarrow E \quad E \rightarrow B, C$$

Now we create one relation for each functional dependency:

$$R_1(A, B, C) \quad R_2(A, D, E) \quad R_3(B, C, D, E) \quad R_4(B, C, E)$$

We need to check that one relation contains a key of R. Indeed, $\{A, D, E\}$ is a key of R, so we do not need to add a fresh relation.

Finally, we remove relations that are subsumed by other relations:

- $\{B, C, E\} \subseteq \{B, C, D, E\}$, so we remove R_4 .

The third normal form is:

$R_1(A, B, C)$

$R_2(A, D, E)$

$R_3(B, C, D, E)$

6. Consider the transformation to BCNF and 3NF above. What functional dependencies have been lost in these transformations?

Solution: The following functional dependencies have been lost in the BCNF transformation:

$B, C, D \rightarrow E$

$A, E \rightarrow D$

No functional dependencies have been lost in the transformation to 3NF.